

DEMO-TI64-001

Ti-64 · disc
Machinability disc



ID 4556298d

ACTIVE

OWNER Dennis Premoli dpremoli1@sheffield.ac.uk	STATUS active	MANUFACTURED 15/03/2024	ENTRY CREATED 15/03/2024
LOCATION D1 Lab, Sir Robert Hadfield Building, Sheffield	SURFACE FINISH Machined	FORM disc	MASS 480 g
DIMENSIONS Ø80 × L15 mm	DENSITY 4.41 g/cm ³	MATERIAL Ti-64 AA	PROJECT DEMO-PA Ti-64 FAST Billet Study

SAMPLE LIFE

15/03/2024

Origin

CUT_FROM DEMO-TI64-BILLET 40%

15/03/2024

Sample created

Ti-64 · Machinability disc

01/04/2024 – 04/05/2024

Manufacturing 8

CAMPAIGN Roughing Campaign Finishing Campaign PROJECT DEMO-PA DEMO-PB

OP CODE	OPERATION	VC	FEED	SPINDLE	DOC	DATE
DEMO-TI64-001-R1	Turning · MT-R	20	0.15	1,000	0.5	01/04/2024
DEMO-TI64-001-R2	Turning · MT-R	30	0.15	1,000	0.5	02/04/2024
DEMO-TI64-001-R3	Turning · MT-R	40	0.15	1,000	0.5	03/04/2024
DEMO-TI64-001-R4	Turning · MT-R	60	0.15	1,000	0.5	04/04/2024
DEMO-TI64-001-F1	Turning · MT-F	60	0.05	1,500	0.1	01/05/2024
DEMO-TI64-001-F2	Turning · MT-F	80	0.05	1,500	0.1	02/05/2024
DEMO-TI64-001-F3	Turning · MT-F	100	0.05	1,500	0.1	03/05/2024
DEMO-TI64-001-F4	Turning · MT-F	120	0.05	1,500	0.1	04/05/2024

05/06/2024 – 18/06/2024

Testing 2

DATE	TYPE	OPERATOR	STATUS
05/06/2024	tensile	A. Operator	analysed
18/06/2024	tensile	A. Operator	analysed

05/07/2024 – 12/07/2024

Sectioned into children 2

DEMO-TI64-C1 COUPON 50% 05/07/2024

DEMO-TI64-C2 COUPON 20% 12/07/2024

NOTES

Primary demonstration sample: cut from the FAST billet, turned across a roughing and a finishing campaign, tensile-tested, then sectioned into two child coupons.

